

From Diagnosis to Solution

A Revision Roadmap and Retrieval-Aware Extension for the SLEEP Paper

Research proposal for a main-conference submission

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Professor Guha,

This memo requests your sign-off on a scoped, four-week plan to take *Recognition Without Recall* from its current diagnostic form to a main-conference submission. The plan does two things: it consolidates the review feedback — yours and the IBM AI CTO review — into a single phased schedule with costs, and it adds one substantive contribution that was not in the reviewed draft. That addition, a *retrieval-aware warm-up phase*, is the focus of this memo, because it is what converts the paper from “here is where the architecture breaks” into “here is where it breaks and the first principled step that repairs it.” The rest is execution against feedback you have already given, so I keep it brief and spend the space on what is new.

1. Where the paper stands

The paper is complete, archived, and reviewed:

- **Archived** on Zenodo with a citable DOI: [10.5281/zenodo.20176028](https://doi.org/10.5281/zenodo.20176028).
- **Code and logs** released at github.com/Adineu03/sleep-framework — ~3,500 lines, 297 tests, full experimental logs.
- **Reviewed** by three domain experts across academia and industry. This proposal integrates the actionable items from those reviews.

The findings themselves are unchanged and I will not restate them beyond the three you already weighed: the +0.16 recognition signal ($\sim 2.9\sigma$), the single-cycle Pareto frontier with no operating point at $\text{BCP} < 1.05$ and $\text{DRA} > 0.05$, and the multi-cycle preservation crossover (BCP 2.31 vs. 4.74 against naive LoRA at cycle 3). The open problem the paper names — consolidation, i.e. moving attention-time recognition into weights that support generation — is exactly what the extension below targets.

2. The key addition: a retrieval-aware warm-up phase

Your review (item #5) recommended framing retrieval-aware fine-tuning explicitly as *the next step* rather than vague future work. This proposal takes that recommendation as its centerpiece: instead of a mention in the discussion, we implement it inside the SLEEP pipeline and measure whether it closes the recognition–recall gap on the same 200-fact benchmark.

2.1 Mechanism

The gap has a specific mechanistic cause, established in the paper: a frozen pretrained model can let injected key–value (KV) memory *bias a classification* but cannot let it *steer autoregressive generation*, because the pretrained prior dominates sampling. The warm-up teaches the missing skill — *use the KV bank during generation* — without teaching any test fact:

1. Start from the frozen base model. Inject random KV entries drawn from a *general* corpus (not the 200 facts).
2. Fine-tune a small LoRA adapter (optionally a lightweight gating module) on standard language-modeling loss *with KV memory active*. The model learns to attend to relevant bank content while generating.
3. Freeze the warmed-up component and run SLEEP exactly as before.

The warm-up carries no fact-specific knowledge; it installs a reusable capability. The prediction is that recall rises because the model now knows *how* to use memory at sampling time, not just at comparison time. An optional enhancement — a few-hundred-parameter trainable gate between memory attention and the residual stream, trained during warm-up and frozen thereafter — provides a permanent, learned bridge from recognition to recall.

2.2 Expected result

The extension slots into a new results subsection with a single decisive table, run on the same benchmark and metrics already in the paper:

Condition	DRA	BCP	Status
SLEEP without warm-up	~0.005	~1.00	current result
SLEEP with warm-up	TBD	TBD	the extension
Naive LoRA	0.275	2.94	existing baseline

The success criterion is modest and pre-registered: if warmed-up SLEEP reaches $\text{DRA} \approx 0.10\text{--}0.15$ at $\text{BCP} < 1.5$, the paper demonstrates that the recognition–recall gap is *closable within the SLEEP framework* without abandoning the safety architecture. Even a partial narrowing is publishable, because it converts the central negative result into a diagnosed-and-treated result.

2.3 Why this changes the paper’s standing

The extension completes a full research arc: (1) we build a biologically-faithful consolidation system; (2) we find where it breaks — recognition without recall; (3) we diagnose why — frozen models do not use KV memory during generation; (4) we locate the fix in adjacent, peer-reviewed literature; (5) we apply it inside SLEEP as the warm-up; (6) we show the gap narrows. Theory, implementation, failure, diagnosis, principled fix — the arc that distinguishes a main-track paper from a findings-track one.

3. The extension is grounded, not speculative

Each component of the warm-up has a published precedent, and one of them independently validates SLEEP’s own tagging mechanism:

- **EM-LLM** (Fountas et al., ICLR 2025) segments the KV cache by *surprise* — nearly identical to SLEEP’s tagging layer — for infinite-context retrieval. It validates our tagging design and lets us position SLEEP precisely as “EM-LLM plus consolidation into permanent weights.”
- **G-MemLLM** (arXiv:2602.00015, Jan 2026) pairs a *frozen* backbone with a small trainable gated latent-memory bank and reports a 13.3% accuracy gain on ZsRE for Llama 3.1 — direct evidence that a lightweight trainable component lets a frozen model use external memory. This motivates the optional gating module.
- **MemoryLLM** (Wang et al., ICML 2024) demonstrates a self-updatable transformer memory with a learnable write-gate and top- k router, establishing that gated memory in a transformer is a mature, reviewed idea.
- **Memorizing Transformers** (Wu et al., ICLR 2022) — already cited in the paper as the mechanism that closes the gap — solves it by training on a memory-augmented objective. The warm-up is a lighter-weight, in-pipeline version of exactly that idea.

Positioning SLEEP against EM-LLM and G-MemLLM connects the paper to two recent top-venue works and frames the extension as the natural, expected next step rather than an isolated trick.

4. Consolidated revision roadmap

The eleven review items collapse into three phases. P1 generates the data every later item depends on; P2 makes the results comparable and complete; P3 (which now houses the extension) delivers the polish and the solution narrative.

P1 — Critical (Weeks 1–2, ~\$70–80)	
Multiple random seeds (3–5) across all key experiments; report mean $\pm$ SD	~\$20–30
Extended multi-cycle: 10–20 cycles to settle whether BCP plateaus or climbs	~\$20
Multi-model & multi-precision: add Llama 3 / Mistral, test in float32, probe 1–3B scale	~\$30
P2 — High (Week 3, ~\$30–40)	
Standard benchmark (LAMA / ROME–MEMIT) plus 50 post-cutoff real facts	~\$15
Missing baselines: RAG on the same facts, EWC-only, in-context injection	~\$10–15
Two-stage validation: surprise pre-filter + direct mini-recall gate (attacks the C4 self-grading gap)	~\$5
Tighten biological framing: constrain-or-trim per your item #7	\$0
P3 — Polish & Solution (Week 4, ~\$5–10)	
Retrieval-aware warm-up extension (§2) — elevated from item #8 to a core result	~\$2–4
Failure-mode table mapping C1–C5 to cause, observability, and fix	\$0
Calibration plot: P(correct) for tagged vs. untagged facts	\$0
Repository polish (README, citation block)	done

Totals: eleven items, ~\$100–120 in compute, ~4 weeks of focused work. The extension itself is the cheapest line in the plan (~\$2–4, roughly half a day) and the highest-leverage.

5. Requested decision

I request your sign-off to proceed with the plan as scoped above. Concretely, I am asking for three things:

1. **Approval of the phased scope and sequencing** (P1 $\rightarrow$ P2 $\rightarrow$ P3) on the four-week, ~\$120 budget.
2. **Confirmation that the retrieval-aware warm-up is the right centerpiece** — that elevating item #8 into a full extension is the correct bet for a main-track story, or your steer if you would weight it differently.
3. **Guidance on the target venue and deadline** so I can align the four-week completion with the nearest suitable main-track window (e.g. NeurIPS, ICLR, or an ACL-family venue).

With your go-ahead I will begin P1 immediately; the seed and multi-cycle runs are already scripted and can start the same day. I will report P1 results before committing P2 compute, so you retain a checkpoint after the most decision-relevant data lands.

Respectfully submitted — Aditya Tripathi & Dr. Rachit Garg. Paper, code, and reviews available at the DOI and repository above.

Key references for the extension

- [1] Fountas et al. *Human-inspired Episodic Memory for Infinite-Context LLMs* (EM-LLM). ICLR 2025.
- [2] G-MemLLM: *Gated Latent Memory Augmentation for Long-Context Reasoning in LLMs*. arXiv:2602.00015, January 2026.
- [3] Wang et al. *MemoryLLM: Towards Self-Updatable Large Language Models*. ICML 2024. (See also *M+*, ICML 2025.)
- [4] Cetin et al. *Neural Attention Memory Models* (NAMMs). ICLR 2025.
- [5] Wu et al. *Memorizing Transformers*. ICLR 2022.